SPOTIFY MUSIC DATASET

Motivation

The choice of topic and dataset for this project was the result of a structured group decision-making process. As a first step, each of the four group members independently proposed a broad theme they found interesting. The four proposals were then put to a vote, with each member voting for someone else's suggestion, and the winning theme was health. With this direction in mind, we searched for suitable datasets on Kaggle, choosing four to five dataset candidates that seemed to meet the project requirements. A second vote was held, and the most popular option was a dataset related to sleep health, which included variables such as sleep duration, sleep quality, physical activity level, stress level, BMI category, and the presence of sleep disorders, among others.

The dataset looked interesting and well-structured, but we discovered that the data had been synthetically generated, that is, it was not collected from real individuals but artificially produced to simulate realistic patterns. Since the data had not been collected from real observations, we did not think it would make much sense to draw conclusions from it, so we decided to look for other options… We then evaluated the remaining dataset candidates in order of preference, only to find that they were all synthetic as well.

Faced with this situation, we decided to change direction and take the second most voted theme from the initial round: music. The search process was repeated on Kaggle, but again we had the same problem: the vast majority of music-related datasets available are also synthetically generated. After a more exhaustive search, we eventually found a dataset about Spotify tracks built from real audio features and metadata retrieved via the official Spotify API. We carefully reviewed its variables and confirmed that it satisfied all the technical requirements of the project: it contains a large number of observations, as well as numerous variables to analyse, such as danceability, energy, loudness, speechiness, and acousticness, among others.

Apart from fulfilling the project requirements, we also found this dataset genuinely interesting to work with. Music is something we all engage with on a daily basis, and the idea that characteristics like energy, tempo or danceability can be measured and analysed statistically felt both fun and meaningful to us. We were curious to see whether these numbers could actually tell us something about what makes songs sound the way they do, which is what ultimately convinced us to go with this dataset.

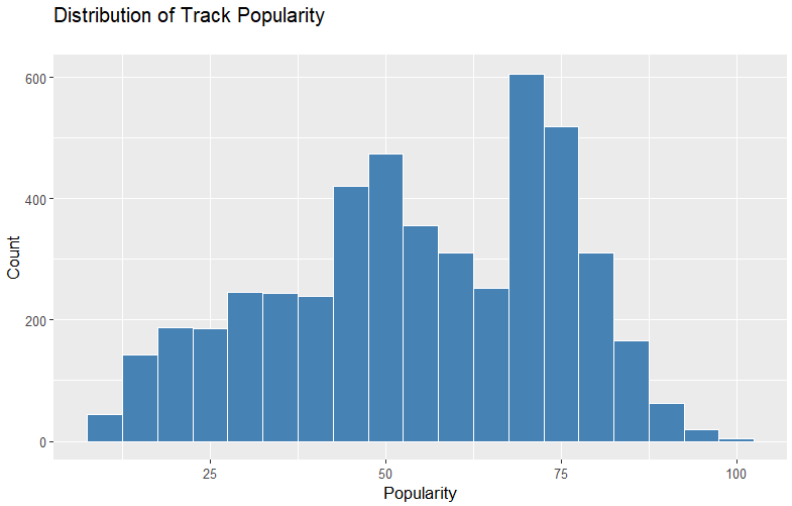
Dataset description

The dataset used in this project was obtained from Kaggle, licensed by the Open Knowledge Foundation, and contains audio features and metadata for 1,686 Spotify tracks, all of them highly popular songs drawn from a variety of playlists. The data was originally retrieved via the official Spotify API, which provides a standardised set of audio features for each track computed by Spotify's internal algorithms.

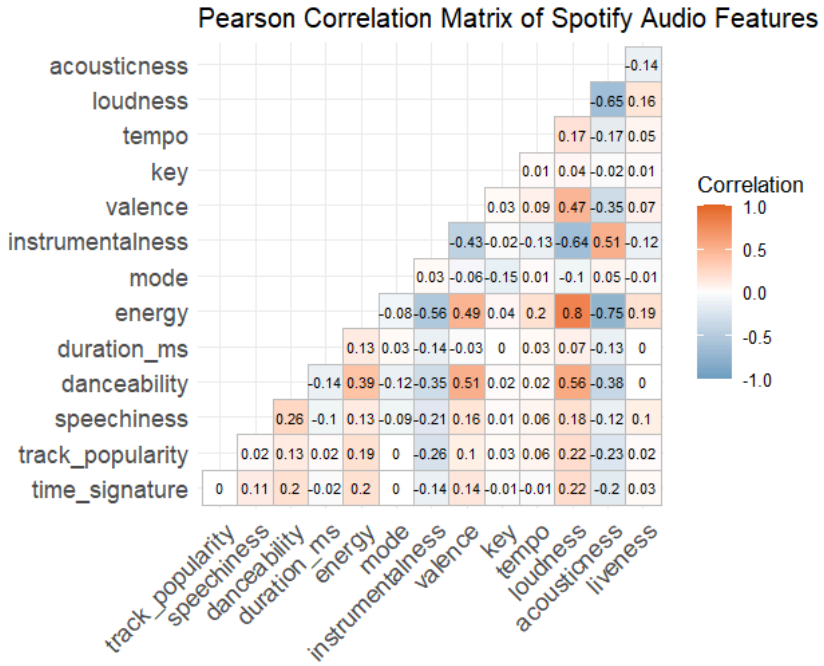
The dataset contains 29 variables in total, which can be grouped into three broad categories. The first group consists of identification and metadata variables, such as the track name, artist, album name, release date, playlist name, and various API-specific identifiers (track ID, URI, href). These variables are not used directly in the statistical analyses but provide useful contextual information. The second group contains the continuous audio feature variables, which are the core of our analysis: energy, tempo, danceability, loudness, liveness, valence, speechiness, instrumentalness,

acousticness, and duration in milliseconds. These variables are all numeric and capture different perceptual and acoustic dimensions of each song. The third group includes categorical variables: playlist genre (10 categories, such as pop, rock, jazz, hip-hop or classical), playlist subgenre (53 categories), key (12 values corresponding to musical pitch classes), time signature (4 possible values), and mode (binary: 0 for minor, 1 for major). Track popularity is also included as a numeric variable ranging from 0 to 100.

Before fitting any models, we conducted an initial descriptive analysis to understand how the variables are distributed and how they relate to one another. Looking at the distribution of track popularity, the data appears roughly normal but exhibits a slight leftward skew, with a prominent spike in tracks scoring between 65 and 75. This clustering at the higher end of the scale is what ultimately informed our decision to set the boundary for highly popular tracks at a score of 68 for the logistic regression analysis.

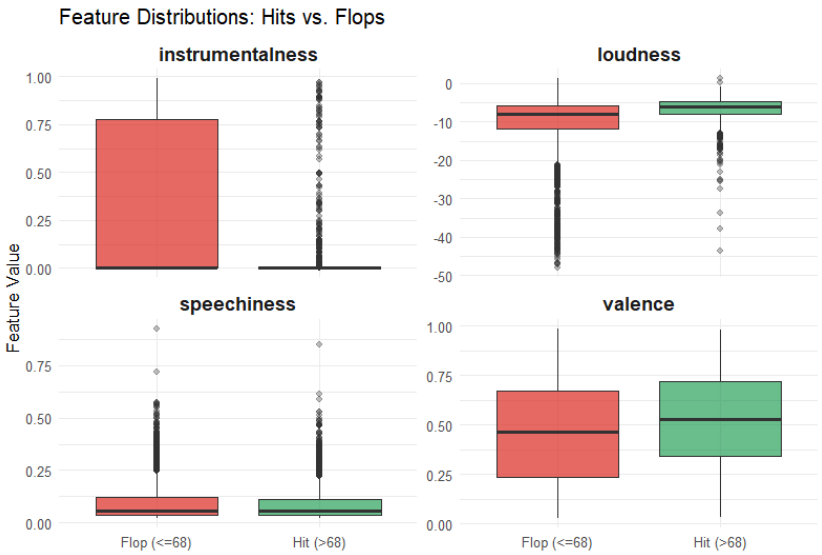


To explore the relationships among the continuous audio features, we computed a Pearson correlation matrix. The matrix revealed several expected acoustic patterns. For instance, loudness and energy are highly positively correlated (0.69), as are valence and danceability (0.51). Conversely, acousticness is strongly negatively correlated with energy (-0.56) and loudness(-0.45).



When looking at our target variable, track popularity does not show any massive linear correlations with individual audio features, though slight positive associations exist with loudness and danceability, and slight negative ones with acousticness and instrumentalness. This indicates that while individual features contribute to a song's success, no single variable can perfectly predict it on its own, which justified our need to build the multiple regression models detailed in the following sections.

Since we anticipated the classification task for the second part of the project, we also grouped the dataset into highly popular tracks (scores > 68) and less popular tracks to visualise how the key features were distributed across these two groups. This revealed some distinct acoustic profiles. For example, the instrumentalness and speechiness of highly popular tracks are heavily compressed near zero, whereas less popular tracks exhibit a much wider variance and higher medians for these features. On the other hand, highly popular tracks generally display higher overall loudness compared to less popular ones. It is worth noting here that Spotify measures loudness in Decibels relative to Full Scale, where 0 represents the absolute digital maximum, which explains why the values in our dataset and graphs are all negative. These visual differences strongly suggested that features like loudness and instrumentalness possess significant discriminative power, foreshadowing the prominent roles they would play in our logistic regression model.



Linear regression

The goal of this section is to build a linear regression model to predict the popularity of a Spotify track (track_popularity) from its audio features. Before fitting any model, the dataset was split into a training set and a test set using a 70/30 partition, with 70% of the observations used to train the model, and the remaining 30% set aside to evaluate the model's performance on unseen data. The split was stratified by track_popularity to ensure that the distribution of the response variable was preserved in both sets. This is a standard practice used to prevent overfitting, which happens when a model works very well on the data it was trained on but gives poor predictions when applied to new data.

Only numeric variables were kept as candidate predictors, excluding all identification and metadata columns (track name, artist, album, playlist identifiers, etc.) that carry no statistical information about a song's popularity. The considered variables were: energy, tempo, danceability, loudness,

liveness, valence, speechiness, instrumentalness, acousticness, duration_ms, key and time_signature.

We started by fitting a full model including all 13 predictors. The full model was globally significant (F-statistic: $p < 2.2e-16$) with an adjusted R^2 of 0.086, but many individual predictors were not significant, such as danceability, energy, mode, key, liveness, loudness or tempo, whose p-values were above 0.05. We then applied backward selection using the *stepAIC* function from the MASS package in R, which iteratively removes the variable whose elimination produces the greatest improvement in AIC. After removing the non-contributing variables, the final model kept 7 predictors: speechiness, energy, instrumentalness, valence, loudness, acousticness, and liveness.

Looking at the final model, the most influential variable by far is instrumentalness, with a coefficient of -12.17, which means that tracks with a high portion of instrumental content tend to be considerably less popular. Acousticness also has a strong negative effect, since it has a -9.49 coefficient, suggesting that the more acoustic songs are generally less popular in this dataset. Speechiness and Valence, which have -7.69 and -3.56 coefficients, respectively, also show negative relationships with popularity, meaning that tracks with more spoken words or a more positive or happy sound tend to score lower in popularity.

Before finalising the linear model, it was necessary to ensure the stability of the estimated coefficients by testing for multicollinearity. We calculated the Variance Inflation Factor (VIF) for all predictors kept in the backwards-selected model. All variables yielded VIF values well below the empirical threshold of 5, indicating that the predictors are sufficiently independent and the model does not suffer from significant multicollinearity.

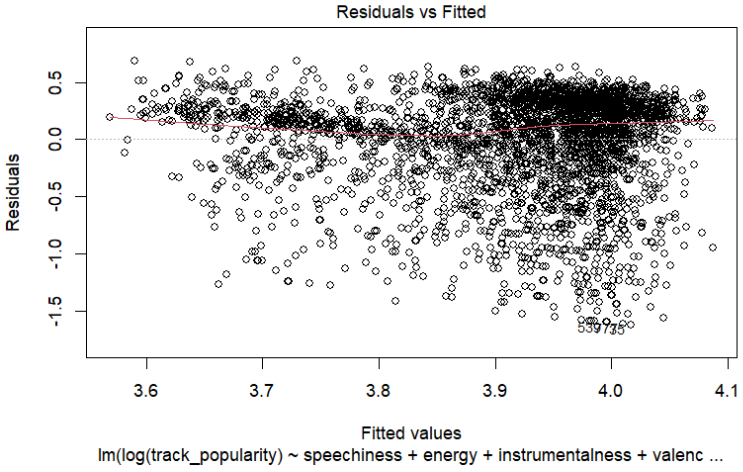
To explore whether the relationships between audio features and popularity were non-linear or dependent on one another, we built a second, more complex model. This model included a polynomial term for energy and an interaction term between acousticness and energy. We then compared this complex model to our simpler backwards-selected model using an Analysis of Variance (ANOVA) test. The test returned a highly significant p-value ($p < 0.001$), technically indicating that the complex model was a statistical improvement. However, upon closer inspection, the complex model only reduced our Residual Sum of Squares by about 0.4%. Because our sample size is so large (over 3,300 observations), the test flagged this microscopic improvement as mathematically significant. We ultimately decided to reject the complex model based on the principle of parsimony. That negligible 0.4% gain in predictive power was not worth the severe loss in interpretability, as adding those interaction terms caused multicollinearity and made it impossible to explain the variables' individual effects straightforwardly.

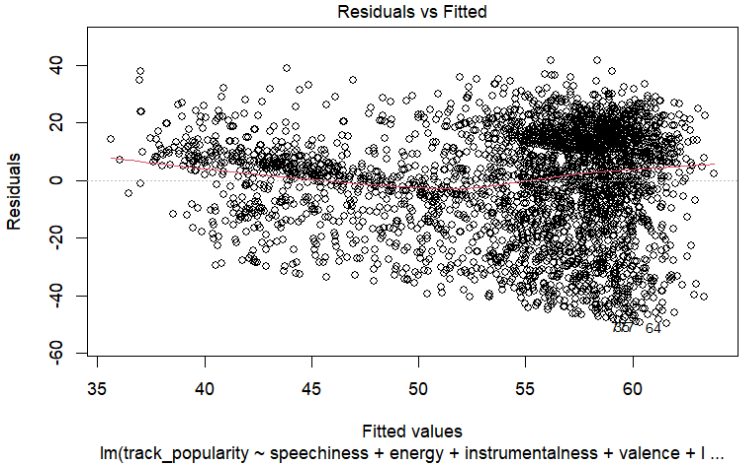
Once we had our final model, we needed to check if it actually met the basic assumptions of Ordinary Least Squares (OLS) regression. We looked at the Residuals vs Fitted plot to check for homoscedasticity and linearity, and it became evident that the constant variance assumption was violated. Instead of a random, even spread, the residuals actually increased as the fitted values grew, creating a downward funnel shape.

We also noticed a harsh diagonal line at the top of the plot, acting like a ceiling that blocked positive residuals. This non-random shape actually comes from the way Spotify scores track popularity. Since the score is strictly capped at 100, the math forces this boundary. Because a residual is just the actual value minus the predicted value $(e_i) = (y_i) - (hat_y)$, the biggest possible positive error for any prediction is $Max (e_i) = 100 - (hat_y)$. So, if our model predicts a popularity of 60, the error could theoretically be +40. But if it predicts a 95, the error can't be more

than +5. This "ceiling" naturally squishes the variance as predictions get higher, which inherently breaks the OLS assumptions.

To address this heteroscedasticity, we applied a logarithmic transformation to the target variable, which is a standard approach. While the logarithm changed the overall scale, the strict diagonal boundary remained evident. Additionally, the log transformation caused a severe drop in the model's predictive power, reducing the adjusted R^2 from approximately 0.09 to 0.05, and causing previously meaningful predictors, such as speechiness, to completely lose their statistical significance.





Seeing this, we decided the log transformation wasn't a real fix for the boundary issue and only made the model harder to interpret. We chose to stick with our original backward-selected model, but we acknowledge this ceiling effect as an inherent limitation. It means the standard errors for our higher predictions might be slightly biased.

Finally, to see how the model actually performs on new data, we tested it on the 30% hold-out test set we created earlier. We used the model to predict the popularity of the unseen tracks and calculated the Root Mean Squared Error (RMSE) and the out-of-sample R^2.

The model got an out-of-sample R^2 of 0.0709. This is very close to the training R^2 (0.089), which is good news because it means our model is robust and not overfitting. However, the RMSE was 19.3. Practically, this means our predictions deviate from the true popularity score by nearly 20 points on average. This high error margin just confirmed what we suspected from our plots: trying to predict an exact popularity score using linear audio features is incredibly noisy.

While the linear regression model provided valuable insights into the relationships between audio features and track popularity, its predictive utility remained compromised by the ceiling of the target variable. Given these limitations, attempting to predict an exact continuous popularity score may not be the optimal approach for this specific dataset.

By splitting the popularity scores into a binary "highly popular" vs "less popular" category, we can use logistic regression instead. Logistic regression naturally handles probabilities between 0 and 1, which completely avoids the ceiling and heteroscedasticity problems we just dealt with. Plus, categorising the data like this fits much better with how the music industry actually works—"Is this going to be a hit, or is it going to flop?" rather than asking for an exact index score. Therefore, the

next section will cover the construction and evaluation of a logistic regression to classify track popularity.

Logistic regression

This section applies logistic regression to predict whether a track is highly popular based on its audio features. Unlike linear regression, which models popularity as a continuous outcome, logistic regression requires a binary target variable. The following subsections describe how this variable was constructed, how the data was prepared, and how the model was built and evaluated.

Our general dataset is a combination of the low and high popularity, which is why we have chosen 68 as the split of popularity. The low dataset goes from 11 to 68, and the high goes from 68 to 100. There is an overlap in 68 which generates duplicated data. This duplicated data is removed by using the full_join function.

The binary variable was defined based on the original dataset split, classifying tracks with popularity > 68 as high popularity (1) and the rest as low popularity (0). That is, 67.4% is labelled as 0 and 32.6% is labelled as 1. There is a moderated imbalance; there are more tracks with low popularity, but it is still acceptable.

In this section, we prepared the data for the logistic regression modelling. First, we reduced the dataset to only the relevant variables, keeping the binary target variable popular and nine continuous audio features. All identifier columns, URLs, and text fields were excluded as they carry no predictive value. Consistent with our methodology in the linear regression analysis, we split the dataset into a training set (70%, n=3350) and a test set (30%, n=1437). We again used stratified sampling on the new binary target variable to ensure the ~ 67/33 class distribution was preserved across both subsets.

A full logistic regression model was first fitted on the training data using all ten predictors. As with the linear regression, backward stepwise selection based on AIC was then applied using the stepAIC function from the MASS package, which iteratively removed the least informative predictors. The final model retained 5 predictors: duration_ms, valence, speechiness, loudness, and instrumentalness. The variables energy, danceability, tempo, and liveness were removed during the selection process, as their inclusion did not improve the AIC of the model.

The odds ratios, obtained by exponentiating the model's coefficients, provide a clear way to interpret the results. Because our predictors were normalized during the data preparation phase, each odds ratio represents the multiplicative change in the odds of a track being highly popular for a one-standard-deviation increase in that feature, holding all other variables constant.

The most influential negative predictor was instrumentalness. With a coefficient of -0.855, its odds ratio is exp(-0.855) = 0.425. This indicates that a one-standard-deviation increase in instrumental content decreases the odds of a track being classified as highly popular by 57.5%, which strongly aligns with our findings from the linear regression section. Speechiness also showed a notable negative association (odds ratio = 0.775, representing a 22.5% decrease in odds), and valence was similarly negative (odds ratio = 0.913).

Conversely, loudness was the strongest positive predictor in the model. With an odds ratio of 1.785, a one-standard-deviation increase in loudness increases the odds of a track being popular

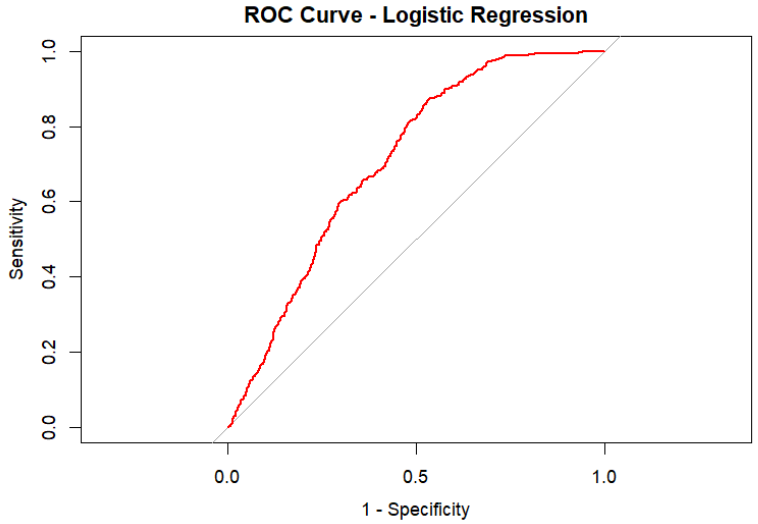
by 78.5%. Finally, the effect of duration_ms was only slightly positive and less impactful (odds ratio = 1.068).

Before accepting the model, multicollinearity was assessed by computing the Variance Inflation Factor (VIF) for each retained predictor. All VIF values were well below the conventional threshold of 5, ranging from 1.023 (duration_ms) to 1.255 (loudness), confirming that the predictors are sufficiently independent and that the coefficient estimates are stable.

The final logistic regression model was evaluated on the held-out test set. Using a classification threshold of 0.5, the predicted probabilities were converted into binary labels and compared against the true values.

The model achieved an overall accuracy of 67.22%, correctly classifying 966 out of 1437 test observations. The sensitivity (true positive rate), that is, the proportion of genuinely popular tracks correctly identified, was 13.65%, while the specificity (true negative rate) was 93.18%. The relatively higher specificity compared to sensitivity reflects the class imbalance in the data: since the majority of tracks are labelled as low popularity, the model tends to be more conservative in predicting high popularity.

To obtain a threshold-independent evaluation of discriminative performance, we computed the Area Under the ROC Curve (AUC). The model achieved an AUC of 0.711, which indicates moderate discriminative ability — meaning the model correctly ranks a randomly chosen popular track above a randomly chosen unpopular one approximately 71.1% of the time.



Given the moderate class imbalance, we also tested an alternative classification threshold of 0.35. Using this lower threshold increased the sensitivity to 78.04% at the cost of reducing specificity to 51.14% and keeping overall accuracy relatively stable at 59.92%. The balanced accuracy improved from 53.41% to 64.59%, suggesting that the optimal threshold may depend on the specific use case — for instance, a lower threshold would be preferable if the goal is to avoid missing potential hits, even at the cost of more false positives.

The logistic regression model confirmed several of the patterns identified in the linear regression analysis, most notably the strong negative association between instrumentalness and popularity, and the similarly negative role of speechiness. By reformulating the problem as a binary classification task, we were able to sidestep the ceiling effect and heteroscedasticity that undermined the linear model.

That said, the model's predictive power remains moderate, with an AUC of 0.711 and an accuracy of 67.22%. This suggests that while audio features do carry some information about a track's commercial success, they are far from sufficient on their own to reliably predict whether a song will become popular. Factors such as artist recognition, promotional activity, playlist placement, and cultural context are likely to play an equally or more important role, but are not captured in this dataset.